Circuit models

11a

When learning a new scientific concept, it can sometimes be difficult to understand what is going on — especially when you cannot actually see it. To help you understand, a simple model can be invented to explain what is happening. However, models do have their limitations and may need to be changed as your understanding becomes better.

In a circuit the power supply—the battery or the powerpack—acts as an energy bank, supplying the electrons with the same amount of energy as they travel. As the electrons move around the circuit, they use some of this energy.

Some materials are more difficult for electrons to move through than other materials. Such materials cause resistance in a circuit. Materials with low resistance allow the current to pass through easily; they are good conductors. Materials with high resistance do not allow much current to flow through; they are resistors. Sometimes the shape and length of the wire affects resistance, for example a thin piece of wire has a higher resistance than a thicker piece of wire.

Below is a model of an electric circuit, called the Human Circuit, which was performed by Ms Smart's class.

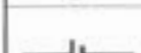
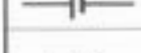


1 Look at the picture above and then complete the following table.

a Match the symbols with their names by joining the appropriate boxes with a line.

b Show which student or activity represents each symbol by joining the appropriate boxes with a line.

c Write the number of the activity in the last column.

Name	Sketch	Student or activity	Activity number
Ammeter		Climbing over a table	
Battery		Counting energy beans	
Globe		Giving out energy beans	
resistor		Pushing through a tube	

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11b

2 Answer these questions about the human circuit model.

a What do the students going around the circle represent? _____

_____

b How would you include a switch in this model? _____

_____

c What represents the amount of voltage supplied? _____

_____

3 In the space below design a **traffic circuit model**. Use any familiar traffic devices, for example cars, traffic lights, signs, roundabouts, etc. Remember to include a battery, switch, some kind of resistance, a globe and an ammeter. Compare your circuit with the rest of the class.

4 In your traffic circuit model what represents:

a the electrons? _____

b the amount of voltage supplied? _____

5 Compare your diagram to the rest of class. Write a list of common features.

_____

_____

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